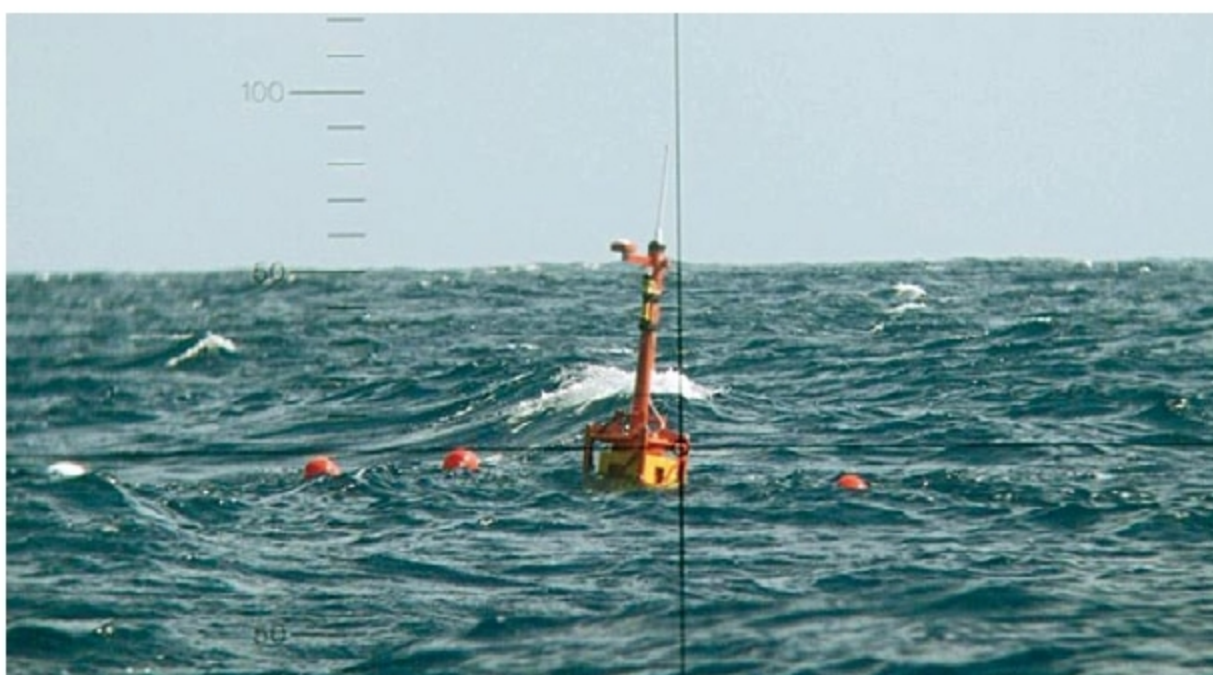


These are then directly transmitted via radio link to naval units under test in real time. imc C-series measurement system can be used for this application. Its batteries allow autonomy for at least twelve hours, and main power can be switched on or off remotely via radio control.

Hydrophone sensors are submerged deep below the buoy and are connected via cabling and waterproof connectors to the imc C-series data acquisition system, sampling at 50kS/s/channel. The system is connected to a GNSS high-performance satellite receiver from JAVAD for accurate position measurements and to synchronise data recording with GPS time. A Wi-Fi modem/router equipped with a booster is used to increase power of the transmitter and omnidirectional antenna.

Today's naval forces require reliable and secure interoperable communications, interception- and jam-resistant voice and data connections in any type of scenario. These need to be designed to maintain and improve the digital sovereignty of customers and make them technologically independent of manufacturers and industrial partners.

Rohde & Schwarz supplies integrated communications and network systems to naval platforms in addition to naval, joint or allied communication networks. Its Navics integrated communications system (ICS) can be used for all onboard



Detecting acoustic emissions of ships and submarines (Credit: www.imc-tm.com)

communications, and Soveron for all external communications. These can be customised for all classes of ships with an all-IP architecture to offer maximum flexibility in design and service.

Soveron stands for high-performance, fully integrated, modular communications solutions for all branches of the armed forces. The portfolio covers portable radios for dismounted soldiers as well as vehicular, stationary and airborne applications and complex systems for ships. One key element is an open source development platform that enables users to develop their own national, independent waveforms. Users can also adopt a suite of network capable, high data and jam-resistant waveforms to cover a wide range of communication scenarios.

Dependable and accurate measurement of pressure, temperature and other variables within various tanks is an important measure in preventing the escape of aggressive fluids, controlling water circulation systems in ship operations. Sensor technology employed here has to meet numerous stringent requirements, above all materials utilised must be robust enough for use over the longer term. Both wastewater and freshwater tank systems are monitored using built-in sensors, which guarantees an optimum water supply across the high seas.

Kistler provides high-precision pressure sensors with long-term stability and easy-to-handle diagnostic equipment for large ship engines. Offline cylinder pressure plays a key part in monitoring two- and four-stroke engines to optimise maintenance intervals, detect damage ahead of time, ensure compliance with emission limits and minimise fuel consumption. Rugged piezoelectric sensors and measuring chains from Kistler deliver high-precision data for continuous operational monitoring of large engines.

With sensor technology, errors are detected early and control of the combustion process is optimised. Sensors that can withstand high operating temperatures are mounted on the indicator valve for periodic measurements. **EFY**



Soveron AR software-defined airborne radio (Credit: www.rohde-schwarz.com)

